

RE-ESTIMATIONS OF THE ZMIJEWSKI AND OHLSON BANKRUPTCY PREDICTION MODELS

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ABSTRACT

Current accounting research uses the Zmijewski (1984) and Ohlson (1980) bankruptcy prediction models as proxies for financial distress/bankruptcy. Such use assumes that the models' predictive powers transcend to time periods, industries, and financial conditions outside of those used to originally develop the models. The objective of this paper is to address whether the construct validity of the financial distress/bankruptcy proxies (based on the original models) used in those recent studies is possibly open to question. The evidence provided in this study suggests that researchers who use the Zmijewski and Ohlson models using recent data should re-estimate the models' coefficients to improve the predictive accuracy of the models.

1. INTRODUCTION AND MOTIVATION

Though the Zmijewski (1984) and Ohlson (1980) models were developed using samples from the 1970s, there is limited evidence addressing the sensitivity of these models to time periods, financial distress situations, and industries outside those of the original samples. Even so, bankruptcy prediction models such as these are still employed in current accounting research to proxy for financial conditions

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of firms from a variety of industries and time periods (e.g. Altman, 1993; Berger, Ofek & Swary, 1996; Carcello, Hermanson & Huss, 1995; Carcello & Neal, 2000; Chen & Church, 1996; Chen & Wei, 1993; Dichev, 1998; Robertson & Mills, 1991; Subramanyan & Wild, 1996). The current use of bankruptcy prediction models by researchers and practitioners assumes that the models' predictive powers transcend to time periods, industries and financial conditions other than those used to originally develop the models. Consequently, the construct validity of the financial distress/bankruptcy proxies (based on the original models) used in those recent studies is possibly open to question. The objective of this paper is to address this construct validity issue by evaluating the research questions listed below.

Recent studies have demonstrated that the predictive accuracy of the Zmijewski (1984), Ohlson (1980), and Altman (1968) bankruptcy prediction models significantly declines when applied to current time periods (e.g. Grice & Dugan, 2001; Grice & Ingram, 2001). Further, the Grice and Ingram study found that improved predictive accuracy can be achieved with Altman's (1968) model by re-estimating the model using samples from periods close to the test period. These prior findings and the continued use of bankruptcy prediction models lead to three research questions considered in this paper: (1) Do the predictive accuracies of the Zmijewski and Ohlson models increase when re-estimated using samples that are closer to the test periods? (2) Are re-estimated Zmijewski and Ohlson models as accurate for predicting bankruptcy of non-industrial firms as they are for predicting industrial firms? (3) Are re-estimated Zmijewski and Ohlson models as accurate for predicting financial distress conditions other than bankruptcy as they are for predicting bankruptcy?

The remainder of this paper is organized as follows. Section 2 discusses Zmijewski's (*X*-score) and Ohlson's (*Y*-score) models and relevant prior research. Section 3 describes the samples and tests employed to evaluate Zmijewski's and Ohlson's models. Section 4 presents the findings, and Section 5 discusses the implications of those findings for users of the Zmijewski and Ohlson models.

2. CONTRIBUTION TO PRIOR RESEARCH

This section summarizes Zmijewski (1984), Ohlson (1980), and other studies that have developed and evaluated bankruptcy prediction models. It explains the contributions of the present study in identifying and resolving the inherent construct validity issues in those earlier studies.

Zmijewski (1984) used financial ratios that measured firm performance, leverage, and liquidity to develop his model. The ratios were not selected on a theoretical basis, but rather on the basis of their performance in prior studies.

Zmijewski's probit model based on 40 bankrupt and 800 non-bankrupt industrial firms from the 1972 to 1978 time period was:

$$X = -4.3 - 4.5X_1 + 5.7X_2 - 0.004X_3, \quad (1)$$

where X_1 = net income/total assets, X_2 = total debt/total assets, X_3 = current assets/current liabilities, and X = overall index. The score is used to determine the probability of membership in the bankrupt group based on a cumulative normal probability function. Zmijewski (1984) developed numerous models using 40 bankrupt and 40–800 non-bankrupt firms; however, the model based on the 40:800 proportion of bankrupt to non-bankrupt firms is the model most commonly used by accounting researchers (e.g. Carcello & Neal, 2000; Carcello et al., 1995; Chen & Wei, 1993).

Ohlson (1980) indicated that the nine predictors used to develop his model were selected because they appeared to be the ones most frequently mentioned in the literature. He employed logistic analysis to derive his bankruptcy prediction model using nine measures of firms' size, leverage, liquidity, and performance. Based on a sample that included 105 bankrupt and 2,058 non-bankrupt industrial firms from the 1970 to 1976 time period, his model was:

$$Y = -1.3 - 0.4X_1 + 6.0X_2 - 1.4X_3 + 0.1X_4 - 2.4X_5 - 1.8X_6 \\ + 0.3X_7 - 1.7X_8 - 0.5X_9, \quad (2)$$

where X_1 = log (total assets/GNP price-level index), X_2 = total liabilities/total assets, X_3 = working capital/total assets, X_4 = current liabilities/current assets, X_5 = one if total liabilities exceed total assets, zero otherwise, X_6 = net income/total assets, X_7 = funds provided by operations/total liabilities, X_8 = one if net income was negative for the last two years, zero otherwise, X_9 = measure of change in net income,¹ and Y = overall index. The overall index is used to determine the probability of membership in the bankrupt group based on a logistic function.

Though the Zmijewski and Ohlson models are the only models that are re-estimated in this study, the findings of this study may apply to other models that were derived using a similar methodological approach. Studies that have developed and tested bankruptcy prediction models include Altman (1968), Deakin (1972), Mensah (1983), and Zavgren (1985). These studies selected estimation and hold-out samples from different time periods and reported relatively high accuracy rates for the models using small samples and short windows of time. For example, Altman reported that his model exhibited an 84% accuracy rate using a hold-out sample from a time period different from that employed to originally develop the model. It should be noted that the extant literature includes other

studies that also developed and tested models; however, they used estimation² and hold-out samples drawn from the same time period (e.g. [Blum, 1974](#); [Gentry, Newbold & Whitford, 1985](#); [Zmijewski, 1984](#)).

The hold-out sample tests represent more rigorous tests of the models' accuracy. Though lower than estimation sample rates, the hold-out sample accuracy rates reported in prior studies are potentially upwardly biased (meaning the hold-out sample accuracy rates are higher than the rates users should expect when they apply the models) for three reasons: (1) the time periods for the estimation and hold-out sample are not substantially different; (2) the hold-out sample consisted of firms from the same restricted set of industries as those in the estimation sample; and (3) the hold-out samples were small (the largest sample was 111 firms).

Applying the original coefficients of the Zmijewski and Ohlson models to recent samples tests the stability of the relation between bankruptcy and the respective models' financial ratios. [Begley, Ming and Watts \(1997\)](#) applied Ohlson's original model to 1,365 industrial firms and reported a 98% classification accuracy. [Grice and Dugan \(2001\)](#) applied Ohlson's original model to firms drawn from the 1988 to 1991 (1992 to 1999) time period and reported a classification accuracy of 39.8% (34.8%). They also applied Zmijewski's original model to 1988–1991 (1992–1999) firms and reported a classification accuracy of 81.3% (77.6%). This study assesses the construct validity of the models' use as proxies for financial distress/bankruptcy by re-estimating the coefficients using recent firm data. Tests in the current study compare both the predictive accuracy rates and estimated coefficients of the original Zmijewski and Ohlson models to those of the re-estimated models. [Begley et al. \(1997\)](#) re-estimated Ohlson's model using 100 bankrupt and 2,000 non-bankrupt companies from 1980 to 1989. They reported that the re-estimated [Ohlson \(1980\)](#) model exhibited an overall estimation sample classification accuracy of 99%; however, the coefficients of the model changed when re-estimated using data drawn from the 1980s.

Re-estimating the coefficients of models using recent samples evaluates the models in periods that are likely to exhibit economic differences from the periods in which the models were originally developed. [Platt and Platt \(1990\)](#) indicated that differences in the economic environment may change the: (1) relationships between the dependent (e.g. bankruptcy) and independent variables (e.g. financial ratios); (2) average range of the independent variables; and (3) relationships among the independent variables. Platt and Platt suggested that these changes are attributable to shifts in the business cycle, corporate strategy, the competitive nature of the market, and technology.

The hold-out sample tests in prior studies also were potentially biased upward (with respect to a sample of firms from a cross-section of industries) since the hold-out samples consisted of firms from the same industries as those in the estimation

sample. These hold-out sample tests do not provide evidence about the models' predictive power across industries. Tests in the current study evaluate Zmijewski's (1984) and Ohlson's (1980) re-estimated models in industries other than those used to derive the original model. These findings are relevant to accounting researchers who apply bankruptcy prediction models to firms from various industries. Recent examples include (Carcello & Neal, 2000; Carcello et al., 1995; Chen & Church, 1996; Chen & Wei, 1993; Dichev, 1998; Subramanyan & Wild, 1996). Each of these studies assumed the bankruptcy prediction models were valid across industries and time periods other than those used to develop the model. Importantly, Robertson and Mills (1991) indicated that it is not appropriate for models derived for one industry group to be used to evaluate the financial conditions of other industry groups; consequently, the predictive powers of the X and Y-score models should decline using industries different from those used to originally develop the models.

2.1. Bankruptcy or Financial Distress?

Even though the Zmijewski (1984) and Ohlson (1980) models were developed to predict the event of bankruptcy, this event is only one of several indicators of financial distress. It is not clear whether these models are specifically useful for identifying firms that are likely to go bankrupt or whether they are more generally a model for identifying firms experiencing financial distress. While firms that experience financial distress are more likely to declare bankruptcy than other firms, most financially distressed firms are not likely to declare bankruptcy. Gilbert, Menon and Schwartz (1990) suggest that financial dimensions that set apart bankrupt and healthy firms are different from those that separate bankrupt and distressed firms. They developed prediction models using both bankrupt/healthy and bankrupt/distressed estimation samples. The model developed using the bankrupt/healthy estimation sample was unable to distinguish failed firms from distressed firms. Bankruptcy usually is a joint result of financial stress and other events that precipitate legal action.

Additional analyses included in this study relate to the re-estimated models' abilities to assess financial distress in a variety of situations as identified by codings on the Compustat Database. Compustat maintains codes for bankruptcy, liquidation, reorganization, S&P ratings for bonds vulnerable to default, and S&P ratings for stocks, all of which may identify firms that are financially distressed. If Ohlson's and Zmijewski's models are better suited for predicting bankruptcy than for predicting other manifestations of financial distress, they may not provide appropriate proxies for some of the applications for which they have been used.

Alternatively, if the models predict financial distress rather than just bankruptcy, care should be used in employing the models to identify bankrupt firms because most distressed firms will not declare bankruptcy.

3. RESEARCH DESIGN

This section describes the methodology employed to re-estimate and test the coefficients of the X and Y -score models. In addition, this section describes the selection criteria used to identify the distressed and non-distressed sample companies.

3.1. *Sample*

The analyses in this study used an estimation sample and a hold-out sample, with each sample including distressed and non-distressed firms.³ Distressed companies were defined as those reported by Compustat as meeting one or more of the following conditions: (1) Chapter 11 bankruptcy; (2) Chapter 7 liquidation; (3) bonds vulnerable to default; or (4) low stock ratings. The non-distressed firms were selected randomly from the population of firms that were evaluated by S&P but did not receive poor S&P stock or bond ratings.⁴ That is, companies that maintained codes for data item 280 (282) that were less than 19 (18) were included in the non-distressed population.⁵

The final estimation samples for the X and Y -score models included 1,048 companies (181 distressed and 867 non-distressed) and 1,059 companies (153 distressed and 906 non-distressed). These samples were used to re-estimate Zmijewski's (1984) and Ohlson's (1980) original coefficients shown in Eqs (1) and (2). The final hold-out samples for the X and Y -score models included 1,024 firms (183 distressed and 841 non-distressed) and 1,043 firms (154 distressed and 889 non-distressed). These hold-out samples were used to evaluate the predictive accuracies of the original and re-estimated X and Y -score models. The financial ratios described in Eqs (1) and (2) were calculated for each firm in both samples with data from Compustat's Industrial Annual Research file (CIAR) and Compustat's Industrial Annual file (CIA).

Two subsets of the hold-out samples were used in analyses for this study. A subset of the distressed and non-distressed firms from the industries used by Zmijewski and Ohlson to develop their models was used to evaluate the sensitivity of the models to industry classifications. A subset of distressed firms that Compustat identified as bankruptcies was used to evaluate the sensitivity of the models to bankruptcy as opposed to other financial distress situations.

Table 1A. Profile Analysis for Zmijewski Samples.

Sample ^a	Statistic	X_1	X_2	X_3
Estimation				
Non-distressed ($N = 887$)	Mean	0.0451	0.5465	2.2547
	Std. Dev.	0.084	0.193	2.173
	Min	-0.936	0.027	0.383
	Max	0.428	1.941	35.947
Distressed ($N = 181$)	Mean	-3.0791	0.9495	1.3969
	Std. Dev.	0.916	0.846	2.441
	Min	-10.648	0.056	0.006
	Max	0.354	8.555	19.347
	p -Value ^b	0.000	0.000	0.030
Hold-out				
Non-distressed ($N = 841$)	Mean	0.0376	0.5637	2.1789
	Std. Dev.	0.122	0.231	2.380
	Min	-2.290	0.016	0.144
	Max	0.825	3.375	49.612
Distressed ($N = 183$)	Mean	-0.1745	0.8896	3.1177
	Std. Dev.	0.375	0.646	14.329
	Min	-2.447	0.002	0.080
	Max	0.877	5.816	184.104
	p -Value ^b	0.000	0.000	0.056

Note: Variables: X_1 = net income/total assets; X_2 = total liabilities/total assets; X_3 = current assets/current liabilities.

^aThe distressed group includes companies that experienced bankruptcy or liquidation as well as those that received low S&P ratings for their bonds or stock. The non-distressed group includes companies that were rated by S&P and did not receive low bond or stock ratings.

^b p -Value one-tailed t -test of differences in variable means between the distressed and non-distressed groups.

Tables 1A and 1B report descriptive statistics, by distressed and non-distressed groups, for the estimation samples used to re-estimate the X and Y -score models' coefficients. A comparison of the distressed and non-distressed variable means for the Zmijewski estimation sample indicated that the ratios deteriorated in the distressed group. For example, net income to total assets was lower for the distressed (-3.0791) than for the non-distressed group (0.0451). Also, the total debt to total assets and current assets to current liabilities ratios were higher for the distressed than for the non-distressed group. The p -values for the test of mean differences between distressed and non-distressed companies were significant for the net income to total assets, total debt to total assets, and current assets to current liabilities variables. Zmijewski (1984) did not report whether the variable means

Table 1B. Profile Analysis for Ohlson Samples.

Sample ^a	Statistic	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9
Estimation										
Non-distressed ($N = 906$)	Mean	1.6858	0.5541	0.2159	0.6534	0.0099	0.0431	0.2208	0.0496	0.0145
	Std. Dev.	1.845	0.177	0.201	0.561	0.098	0.073	0.239	0.217	0.491
	Min	-3.059	0.032	-1.001	0.056	0.000	-0.611	-0.533	0.000	-1.000
	Max	6.814	1.949	0.841	13.324	1.000	0.326	2.705	1.000	1.000
Distressed ($N = 153$)	Mean	-0.9913	0.8981	0.0671	1.3931	0.2026	-0.2492	-0.1167	0.5359	-0.0659
	Std. Dev.	2.021	0.803	0.461	2.457	0.403	0.474	0.515	0.501	0.664
	Min	-5.138	0.055	-3.041	0.051	0.000	-3.056	-4.623	0.000	-1.000
	Max	5.833	8.555	0.841	18.509	1.000	0.354	1.818	1.000	1.000
	p -Value ^b	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.153
Hold-out										
Non-distressed ($N = 889$)	Mean	1.5224	0.5779	0.2059	0.7031	0.0359	0.0352	0.2264	0.0731	-0.0369
	Std. Dev.	1.942	0.248	0.229	0.574	0.186	0.138	0.343	0.261	0.487
	Min	-4.015	0.045	-2.493	0.047	0.000	-2.299	-1.961	0.000	-1.000
	Max	6.335	3.375	0.914	8.023	1.000	0.825	4.917	1.000	1.000
Distressed ($N = 154$)	Mean	-0.8146	0.8649	0.0446	1.2669	0.2468	-0.1789	-0.0871	0.4741	-0.1646
	Std. Dev.	2.036	0.514	0.505	1.766	0.432	0.361	0.573	0.501	0.691
	Min	-4.874	0.084	-3.441	0.097	0.000	-2.447	-4.201	0.000	-1.000
	Max	4.069	4.026	0.861	12.455	1.000	0.877	2.466	1.000	1.000
	p -Value ^b	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Note: Variables: X_1 = log (total assets/GNP price-level index); X_2 = total liabilities/total assets; X_3 = working capital/total assets; X_4 = current liabilities/current assets; X_5 = one if total liabilities exceed total assets, zero otherwise; X_6 = net income/total assets; X_7 = funds provided by operations/total liabilities; X_8 = one if net income was negative for the last two years, zero otherwise; X_9 = measure of change in net income.

^aThe distressed group includes companies that experienced bankruptcy or liquidation as well as those that received low S&P ratings for their bonds or stock. The non-distressed group includes companies that were rated by S&P and did not receive low bond or stock ratings.

^b p -Value one-tailed t -test of differences in variable means between the distressed and non-distressed groups.

for his estimation sample were significantly different between the distressed and non-distressed groups.

A comparison of the distressed and non-distressed variable means for the Ohlson estimation sample also indicated that each variable deteriorated in the distressed group. For example, total liabilities to total assets was higher for the distressed (0.8981) than for the non-distressed group (0.5541). Also, the performance measure, return on assets, was lower for the distressed (-0.2492) than the non-distressed group (0.0431). The p -values for the test of mean differences between distressed and non-distressed companies were significant for each of the variables except for the measure of change in net income. The descriptive statistics for Ohlson's variables using the estimation sample were similar to those reported by Ohlson (1980). Ohlson indicated that all the variable means were significantly different between the distressed and non-distressed groups at the 0.05 level. The descriptive statistics for the hold-out samples are also reported in Tables 1A and 1B. For each sample, these statistics were similar to those of the estimation sample except for the measure of change in net income (Ohlson sample). The mean of this ratio was (was not) significantly different between the distressed and non-distressed firms in the hold-out (estimation) sample.

The Zmijewski and Ohlson models used financial ratios that discriminated among industrial firms. This study evaluated the predictive accuracy and re-estimated the coefficients of the models using both industrial and non-industrial companies. Consequently, the financial data necessary to calculate the models' ratios were not on CIAR and CIA for some non-industrial companies.⁶ Companies were deleted from the sample if CIAR and CIA did not report the necessary financial data. The estimation (hold-out) sample used to test the Zmijewski model included 282 (285) non-industrial and 886 (739) industrial companies. The estimation (hold-out) sample used to test the Ohlson model included 264 (304) non-industrial and 795 (739) industrial companies. Both samples included approximately the same number of industrial and non-industrial firms in the distressed and non-distressed groups. The equal distribution of the industrial and non-industrial firms within each sample was necessary to clearly demonstrate the sensitivity of the X and Y -score models to industry classifications.

Analyses in this study also evaluated the predictive accuracy and re-estimated the coefficients of X and Y -score models using only bankruptcies from the samples. Specifically, the distributions for the distressed firms were partitioned into two categories: (1) those identified as distressed because of bankruptcy; and (2) those identified as distressed for reasons other than bankruptcy. The estimation (hold-out) sample included 70–98 (88–121) bankrupt companies and 83 (62–66) companies that were identified as financially distressed because of reasons other than bankruptcy. The mix of financially distressed companies attributable to

bankruptcy and those attributable to factors other than bankruptcy was used to test the sensitivity of the models to various financial conditions to evaluate the construct validity issue.

3.1.1. Model Re-estimations

The *X* and *Y*-score models' coefficients were re-estimated using the estimation sample. The coefficients were re-estimated using the methodology originally employed to derive the models. Zmijewski and Ohlson used probit and logit analysis to derive the models shown in Eqs (1) and (2). These statistical methodologies were used to re-estimate the *X* and *Y*-score models' coefficients. For each model, the coefficients were re-estimated using: (1) the full estimation sample; (2) a subset of the sample containing only bankrupt firms in the distressed group; and (3) a subset of the sample containing only industrial firms in both distressed and non-distressed groups. Subsequent discussion refers to these models as: (1) the full *X* and *Y*-score models; (2) the bankruptcy-only *X* and *Y*-score models; and (3) the industrial-only *X* and *Y*-score models.

This study compared the magnitude and significance of the coefficients for the industrial-only and bankruptcy-only *X* and *Y*-score models to those for the full *X* and *Y*-score models, respectively, to evaluate whether re-estimations of the models were sensitive to industry classifications or financial conditions. As discussed above, the samples used to re-estimate these models differed in terms of industry or financial conditions. As such, differences in the coefficients of the models would indicate a sensitivity to industry or financial distress situations. This study also compared the coefficients of the full *X* and *Y*-score models to those shown in Eqs (1) and (2), respectively, to provide further evidence about the stability of the models. If the models are stable, then the coefficients reported in Eqs (1) and (2) should be similar to those of the re-estimated models.

The predictive accuracies also were evaluated for the full, bankruptcy-only, and industrial-only (*X* and *Y*-score) models using the hold-out sample. This study used binomial tests to compare the accuracies of the re-estimated models to those using the original Zmijewski (1984) and Ohlson (1980) models reported in the Grice and Dugan (2001) study. These tests provided evidence about whether the predictive power of the *X* and *Y*-score models was affected when they were re-estimated using a recent sample.

4. RESULTS AND DISCUSSION

This section reports the findings of the tests used to evaluate the construct validity of the proxies generated by the *X* and *Y*-score models. The predictive accuracies of the re-estimated models using the hold-out samples and the stability of their

coefficients when re-estimated using the estimation samples are discussed. Evidence related to the models' sensitivity to non-industrial firms and financial conditions is reported.

4.1. Unstable Coefficients

Evidence related to the stability of the X and Y -score models was evaluated by re-estimating the models' coefficients using the estimation samples. If the models are stable, then the coefficients of the original models should be similar to those derived from the estimation samples. As previously discussed, the full estimation sample, a subset of the sample containing only bankrupt firms in the distressed group, and a subset of the sample containing only industrial firms in both distressed and non-distressed groups were used to re-estimate three models: (1) the full X and Y -score models; (2) the bankruptcy-only X and Y -score models; and (3) the industrial-only X and Y -score models. These three samples were used to evaluate whether Zmijewski's (1984) and Ohlson's (1980) models were sensitive to industry classifications or financial condition.

The results reported in Table 2 indicated similar coefficients for the full, bankruptcy-only, and industrial-only X -score models. This finding suggests that

Table 2. Coefficients for Zmijewski's (1984) and Re-estimated Models.

Statistic	Zmijewski's (1984) Model ^a	Full Model ^b	Bankruptcy-Only Model ^c	Industrial-Only Model ^d
Net income/total assets (p -value)	-3.599 (<0.05)*	-4.341 (0.000)*	-4.076 (0.000)*	-4.325 (0.000)*
Total debt/total assets (p -value)	5.406 (<0.05)*	2.106 (0.000)*	1.921 (0.000)*	2.194 (0.000)*
Current assets/current liabilities (p -value)	-0.100 (>0.05)	0.092 (0.024)*	0.991 (0.003)*	0.077 (0.116)

Note: X_1 = net income/total assets; X_2 = total debt/total assets; X_3 = current assets/current liabilities.
 p -Value represents the significance of the coefficient.

^aThese are the coefficients and significance levels reported in Zmijewski's (1984) study. $N = 840$ (40 bankrupt and 800 non-bankrupt firms).

^bCoefficients estimated using the full estimation sample containing all industry classifications and financially distressed companies. $N = 1,048$ (181 distressed and 867 non-distressed companies).

^cCoefficients estimated using a subset of the estimation sample that only includes bankrupt companies in the distressed group. $N = 990$ (123 distressed and 867 non-distressed firms).

^dCoefficients estimated using a subset of the estimation sample that only includes industrial companies. $N = 791$ (155 distressed and 636 non-distressed firms).

*Indicates significance at 0.05 level.

Zmijewski's model was not sensitive to various distress situations and industry classifications. The coefficients that were significant in both the original and full X-score models were lower in the full model than they were in the original model. These variables include net income to total assets and total debt to total assets. Also, the current assets to current liabilities variable was significant in the full X-score model but not in the original model. Thus, the results indicate that the coefficients of the original X-score model are not stable across time periods.

The results reported in [Table 3](#) indicated similar coefficients for the variables that were significant in both the full and bankruptcy-only Y-score models. These variables included log of total assets to price-level index,⁷ total liabilities to total assets, and funds provided by operations to total liabilities. Again, these results suggest that the model was not sensitive to various distress situations even though Ohlson used only bankrupt firms to develop the original model. The coefficients for total liabilities to total assets, funds provided by operations to total liabilities, and dummy variable for negative net income for the industrial-only Y-score model were lower than those for the full Y-score model. Additionally, the coefficient for the net income to total assets variable was significant in the industrial-only Y-score model, but not in the full Y-score model. This finding indicates that Ohlson's model is sensitive to industry classifications. Since the full, bankruptcy-only, and industrial-only Y-score models reported in [Table 4](#) were similar relative to the original Y-score model, the following comparisons for Ohlson's model refer only to the full model.

The coefficients that were significant in both the original and full Y-score models were higher in the original than they were in the full Y-score model. These variables include log of total assets to price-level index, total liabilities to total assets, and funds provided by operations to total liabilities. Additionally, the dummy variable for negative net income was (was not) significant in the full (original) Y-score model. Ohlson also reported that the coefficients for the dummy variable for total liabilities greater than total assets and the measure of change in net income were significant in the original Y-score model; however, these coefficients were not significant in the full Y-score model. The findings provide further evidence that the original Y-score model's coefficients are not stable across time. The differing coefficients and related significance of the [Ohlson \(1980\)](#) and full models indicate that the relationships from period to period between Ohlson's ratios and financial distress change.

Predictive Accuracy for Re-estimated Models: Additional evidence of the stability of the X-score and Y-score models was obtained by applying the re-estimated models to the hold-out samples. [Table 4](#) reports the predictive accuracies of the re-estimated coefficients for the distressed and non-distressed groups for the full, bankruptcy-only, and industrial-only X and Y-score models using the hold-out

Table 3. Coefficients for Ohlson's (1980) and Re-estimated Models.

Statistic	Ohlson's (1980) Model ^a	Full Model ^b	Bankruptcy-Only Model ^c	Industrial-Only Model ^d
Log (total assets/price-level index) (<i>p</i> -value)	-0.407 (<0.05)*	-0.777 (0.000)*	-0.881 (0.000)*	-0.706 (0.000)*
Total liabilities/total assets (<i>p</i> -value)	6.030 (<0.05)*	3.224 (0.000)*	3.931 (0.000)*	2.204 (0.003)*
Working capital/total assets (<i>p</i> -value)	-1.430 (>0.05)	-0.323 (0.323)	0.054 (0.962)	-1.250 (0.075)
Current liabilities/current assets (<i>p</i> -value)	0.076 (>0.05)	0.589 (0.199)	0.166 (0.657)	0.455 (0.300)
1 if total liabilities exceed total assets, 0 otherwise (<i>p</i> -value)	-1.720 (<0.05)*	0.041 (0.761)	0.645 (0.493)	0.553 (0.552)
Net income/total assets (<i>p</i> -value)	-2.370 (>0.05)	-2.810 (0.158)	-0.548 (0.729)	-3.790 (0.106)*
Funds provided by operations/liabilities (<i>p</i> -value)	-1.83 (<0.05)*	-2.854 (0.003)*	-2.886 (0.007)*	-4.591 (0.000)*
1 if net income negative past 2 years, 0 otherwise (<i>p</i> -value)	0.285 (>0.05)	0.372 (0.003)*	0.656 (0.151)	0.157 (0.003)*
Measure of change in net income (<i>p</i> -value)	-0.521 (<0.05)*	0.206 (0.354)	-0.300 (0.278)	0.309 (0.250)

Note: X_1 = log (total assets/GNP price-level index); X_2 = total liabilities/total assets; X_3 = working capital/total assets; X_4 = current liabilities/current assets; X_5 = one if total liabilities exceed total assets, zero otherwise; X_6 = net income/total assets; X_7 = funds provided by operations/total liabilities; X_8 = one if net income was negative for the last two years, zero otherwise; X_9 = measure of change in net income. *p*-Value represents the significance of the coefficient.

^a These are the coefficients and significance levels reported in Ohlson's (1980) study. $N = 2,163$ (105 bankrupt and 2,058 non-bankrupt firms).

^b Coefficients estimated using the full estimation sample containing all industry classifications and financially distressed companies. $N = 1,004$ (153 distressed and 851 non-distressed companies).

^c Coefficients estimated using a subset of the estimation sample that only includes bankrupt companies in the distressed group. $N = 953$ (102 distressed and 851 non-distressed firms).

^d Coefficients estimated using a subset of the estimation sample that only includes industrial companies. $N = 750$ (130 distressed and 620 non-distressed firms).

* Indicates significance at 0.05 level.

Table 4. Comparisons of the Classification Accuracy of the Hold-Out Samples Using Zmijewski's and Ohlson's Coefficients and Those from the Re-estimated Models.

Model	Statistic	Overall	Distressed Group	Non-distressed Group
Zmijewski				
Zmijewski (1984) ^a	Accuracy ^b	81.3%	58.7%	86.1%
Full ^c	Accuracy ^b	85.7%	36.4%	96.3%
	Test statistic ^d	2.577*	4.341*	6.171*
Bankruptcy-only ^e	Accuracy ^b	86.1%	33.2%	97.3%
	Test statistic ^d	2.801*	4.979*	6.793*
Industrial-only ^f	Accuracy ^b	86.1%	41.3%	95.6%
	Test statistic ^d	2.801*	3.387*	5.477*
Ohlson				
Ohlson (1980) ^a	Accuracy ^b	39.8%	95.4%	30.1%
Full ^c	Accuracy ^b	88.7%	59.1%	93.7%
	Test statistic ^d	22.827*	15.319*	29.332*
Bankruptcy-only ^e	Accuracy ^b	88.5%	51.9%	94.8%
	Test statistic ^d	22.648*	18.327*	14.356*
Industrial-only ^f	Accuracy ^b	88.1%	62.9%	92.9%
	Test statistic ^d	22.603*	13.677*	28.971*

^a Zmijewski's (1984) and Ohlson's (1980) models are represented in Eqs (1) and (2).

^b Accuracy rates represent the correct classifications for each model using the hold-out samples.

^c Model estimated using the full estimation sample containing all industry classifications and financially distressed firms. See coefficients in Tables 2 and 3.

^d z-Statistic comparing the re-estimated model's accuracy rates to those of Zmijewski's or Ohlson's original model as reported in Grice and Dugan (2001).

^e Model estimated using a subset of the estimation sample that only includes the bankrupt firms from the distressed group. See coefficients in Tables 2 and 3.

^f Model estimated using a subset of the estimation sample that only includes industrial firms. See coefficients in Tables 2 and 3.

*The null hypothesis of equal accuracy rates is rejected at 0.05 level.

sample. The overall accuracies for the *X* and *Y*-score models range from 85.7 to 86.1% and 88.1 to 88.7%. The results of binomial tests indicate that the overall predictive accuracies for Zmijewski's (81.3%) and Ohlson's (39.8%) original models when applied to the hold-out samples were significantly less than those for the full, bankruptcy-only, and industrial-only *X* and *Y*-score models.⁸ Thus, these results suggest that those individuals who employ the models using recent data should re-estimate the models' coefficients to obtain more construct valid results.

Predictive accuracy of the bankruptcy-only re-estimated models: [Table 4](#) reports the predictive accuracies for the bankruptcy-only re-estimated models using the hold-out sample. The overall accuracies of the bankruptcy-only *X*-score and *Y*-score re-estimated models were 86.1 and 88.5%. These accuracy rates are similar to those using the full models, which indicates that the models are not sensitive to financial distress conditions. These results suggest that though the models were developed to predict bankruptcy, they seem more generally useful for identifying financial distress. This finding should be carefully considered by researchers using model predictions as proxies for financial health, as opposed to proxies for just bankruptcy.

Predictive accuracy of the industrial-only re-estimated models: [Table 4](#) also reports the predictive accuracies for the industrial-only re-estimated models using the hold-out sample. The overall accuracies of the industrial-only *X*-score and *Y*-score re-estimated models were 86.1 and 88.1%. Consistent with the findings for the bankruptcy-only models, these accuracy rates are similar to those using the full models which indicates that the models are not sensitive to industry classification. However, as previously mentioned, researchers should consider that no theoretical justification exists for applying models to firms that are outside the industry classifications used to develop the models.

5. SUMMARY

This study evaluated the sensitivity of [Zmijewski's \(1984\)](#) and [Ohlson's \(1980\)](#) re-estimated bankruptcy prediction models to samples of distressed and non-distressed companies from time periods, industries, and financial conditions other than those used to develop their models. The findings indicated that the accuracy of the models increased when the coefficients are re-estimated. [Zmijewski \(1984\)](#) and [Ohlson \(1980\)](#) reported 98.2 and 96.4% overall accuracies for their original models using samples from the mid-seventies and before. However, the coefficients of the *X* and *Y*-score models changed dramatically when re-estimated using more recent estimation samples. Thus, it appears the relation between financial ratios and financial distress changes over time. Further, the relative importance of the various ratios in predicting distress conditions was not constant.

The results of this study suggest that the current broad application of the original Zmijewski and Ohlson models raises a construct validity issue. Specifically, evidence in this study suggests that those who employ the *X* and *Y*-score models using recent data should re-estimate the models' coefficients. When the models' coefficients were re-estimated using recent data, their predictive accuracies were significantly higher than those of [Zmijewski's \(1984\)](#) and [Ohlson's \(1980\)](#) original

models when applied to data for the hold-out sample. This study also demonstrated that the re-estimated models were not sensitive to industry classifications and financial distress conditions. Thus, researchers who use the models to predict the event of bankruptcy, not just financial distress, should do so cautiously.

NOTES

1. The change in net income was measured as $(NI_t - NI_{t-1})/(|NI_t| + |NI_{t-1}|)$, where NI_t is net income for the most recent period.

2. An estimation sample is the sample used to develop the model. The model is applied to that estimation sample to determine its explanatory power. Estimation sample rates should be high since the firms are classified based on a model estimated using these same firms. This study's focus relates to the predictive power of a model based on a hold-out sample (sample other than that used to develop the model). Prior studies typically rely on the models' predictive powers to address research questions related to time periods other than those used to originally develop the models.

3. This study used S&P ratings for stocks and bonds from Compustat's Industrial Annual Research file (CIAR) and Compustat's Industrial Annual file (CIA) to identify the firms used in this study. CIAR and CIA did not report these ratings prior to 1985. CIAR contains firms that were deleted from CIA for various reasons, including bankruptcy and liquidation. The estimation (hold-out) sample used in this study was from the 1985 to 1987 (1988 to 1991) period.

4. A random number generator was used to select the companies for the non-distressed group. The method used to select the companies was one that: (1) closely equated the number of non-distressed firms in the 1985–1987 and 1988–1991 samples; and (2) minimized the probability of selecting the same firm for multiple years since each sample year included many of the same non-distressed companies.

5. Firms not rated by S&P were excluded from the non-distressed population because it was not reasonable to assume firms were non-distressed just because they were not rated by S&P.

6. CIAR and CIA do not report current assets or current liabilities for financial institutions, life insurance, or property and casualty companies. The industrial classification includes SIC codes less than 4,000 and 5,000–5,999 for the Zmijewski and Ohlson samples.

7. The price-level index ranged from 99.7 (102.3) to 102.3 (116) for the estimation (hold-out) period.

8. The full X and Y -score models were also estimated after outliers were excluded from the samples. Outliers were defined as the upper and lower 1% of the companies based on Zmijewski's and Ohlson's ratios. The results reported in Table 4 did not change when the accuracies of the re-estimated models based on the reduced samples were used.

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